Answer Key - Machine Learning - Undergraduate Level Assessment 20 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) 0; B) 1; C) 2; D) 3
   Note: The null space of the matrix A consists of all vectors that, when multiplied by A, yield the zero vector, and since A has rank 1 (as all rows are linearly dependent), the dimension of the null space, given by the formula dimension = number of columns - rank, is 2 (3 - 1 = 2).

2. **Answer:** A
   Options: A) Lower variance; B) Higher variance; C) Same variance; D) None of the above
   Note: As the number of training examples approaches infinity, the model becomes better at capturing the true underlying patterns in the data, leading to lower variance and improved generalization to unseen data.

3. **Answer:** D
   Options: A) It can not be applied to non-differentiable functions.; B) It can not be applied to non-continuous functions.; C) It is hard to implement.; D) It runs reasonably slow for multiple linear regression.
   Note: Grid search is computationally intensive as it evaluates every combination of hyperparameters across a predefined grid, leading to slow performance, especially with the increased complexity and dimensionality in multiple linear regression scenarios.

4. **Answer:** B
   Options: A) Larger if the error rate is larger.; B) Larger if the error rate is smaller.; C) Smaller if the error rate is smaller.; D) It does not matter.
   Note: A larger number of test examples is needed to achieve statistically significant results when the error rate is smaller because a lower error rate requires more samples to reliably detect the true performance of the model and ensure that the observed results are not due to random chance.

5. **Answer:** C
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: ImageNet indeed contains images of various resolutions, making Statement 1 true, while Statement 2 is false because ImageNet has significantly more images than Caltech-101, which only contains 9,144 images compared to ImageNet's millions.

True / False

6. **Answer:** True
   Options: A) True; B) False

Note: Underfitting occurs when a model is too simplistic to represent the complexity of the training data, leading to poor performance both on the training set and when applied to new, unseen data.

7. **Answer:** True
   Options: A) True; B) False
   Note: Linear hard-margin SVM requires that the data can be perfectly separated by a hyperplane without any misclassifications, which is only possible when the training data are linearly separable.

8. **Answer:** False
   Options: A) True; B) False
   Note: DCGANs (Deep Convolutional Generative Adversarial Networks) do not use self-attention mechanisms; instead, they rely on convolutional layers and batch normalization to improve training stability.

9. **Answer:** False
   Options: A) True; B) False
   Note: Sampling with replacement in bagging reduces overfitting by increasing diversity among the models but does not eliminate it entirely, as individual classifiers can still overfit to their respective training samples.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because calculating P(H|E, F) also requires knowledge of P(E, F|H) and P(E, F) to apply Bayes' theorem correctly.

Long Form Response

11. **Answer:** def get_inv_count(arr, n): | return inv_count
    Note: correct because it outlines the purpose of a function that computes the number of inversions in an array, which are pairs of elements that are out of order, and implicitly suggests the need for a variable to hold the count of these inversions, though the actual implementation details are incomplete.

12. **Answer:** def divSum(n): | return sum; | def areEquivalent(num 1,num 2): | return divSum(num 1) == divSum(num 2);
    Note: The provided function first calculates the sum of the divisors of a given number through the ‘divSum‘ function, and then compares the sums of the divisors of two numbers using the ‘areEquivalent‘ function, correctly addressing the question of whether these sums are equal.

End of Answer Key